HSE+SOC bands of $A_2B_2Te_5$: color = element-orbital, layer thickness = its share

$Pb_2Sb_2Te_5$ direct gap 0.065 eV (A=Pb, B=Sb)

$Pb_2Bi_2Te_5$ indirect gap 0.114 eV (A=Pb, B=Bi)

$Sn_2Sb_2Te_5$ direct gap 0.135 eV (A=Sn, B=Sb)

$Sn_2Bi_2Te_5$ indirect gap 0.137 eV (A=Sn, B=Bi)

E - E_{VBM} (eV)

Γ M K Γ A L H A|LM|H K

each band = stacked layers (inner → outer as listed); layer thickness ∝ share of that element-orbital

A-s (Pb/Sn) B-s (Sb/Bi) Te-s A-p (Pb/Sn) B-p (Sb/Bi) Te-p